



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE	ANALYSIS TIMEFRAME WINDOW
Forensic Uploaded PCAP: sih26_asim_golden.pcap	Forensic Session Duration: 0.00s – 142.08s (...)

INITIATOR GATEWAY

Company Branch Gateway 1

OS: Ubuntu Linux 24.04
Interface: tailscale0 / eth0
Private IP: 100.119.32.83
VPN Daemon: StrongSwan v5.9.8

IPsec ESP

ESP over UDP 4500 (NAT-T) · Tunnel Mode...

RESPONDER GATEWAY

Enterprise Datacenter Core

OS: Ubuntu Linux 24.04
Interface: tailscale0 / eth0
Private IP: 100.127.207.119
VPN Daemon: StrongSwan v5.9.8

TOTAL PACKETS	FLOW VOLUME	AVG RATE	PEAK RATE	INTERFACE MTU	DROPS / ERRORS
557	22.8 MB	4 pkt/s	1,120 pkt/s	1500 bytes	0 (0.00%)
Frames Ingested	Encrypted Payload	Throughput Velocity	Observed Burst	UP / ACTIVE	Monotonic Verified

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

IKE_SA_INIT [COMPLETED / SUCCESS] Initiator/Responder DH public keys and nonces exchanged. Enforced ECP...
IKE_AUTH [COMPLETED / SUCCESS] Mutual authentication executed via Pre-Shared Key. Initial Child SA e...
CREATE_CHILD_SA [MONITORED] Child SA rekeying evaluated. No ephemeral DH transform negotiated (PF...

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

INITIATOR SPI 0x8b14e9f28a1c9034	RESPONDER SPI 0x4a7c10b83f09de21
INBOUND ESP SPI 0xc0a80102	OUTBOUND ESP SPI 0xc0a80103

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE STATUS	BENCHMARK STANDARD
Encryption Algorithm	AES-256-GCM	AES-256-GCM	APPROVED	RFC 8221 / CNSA 2.0
Key Exchange (Diffie-Hellman)	ECP-256	ECP-256	APPROVED	NIST SP 800-77 §4.2
Pseudo-Random Function (PRF)	HMAC-SHA2-256	N/A (AEAD Suite)	APPROVED	RFC 7296
...	...	...	...	NIST SP 800-77



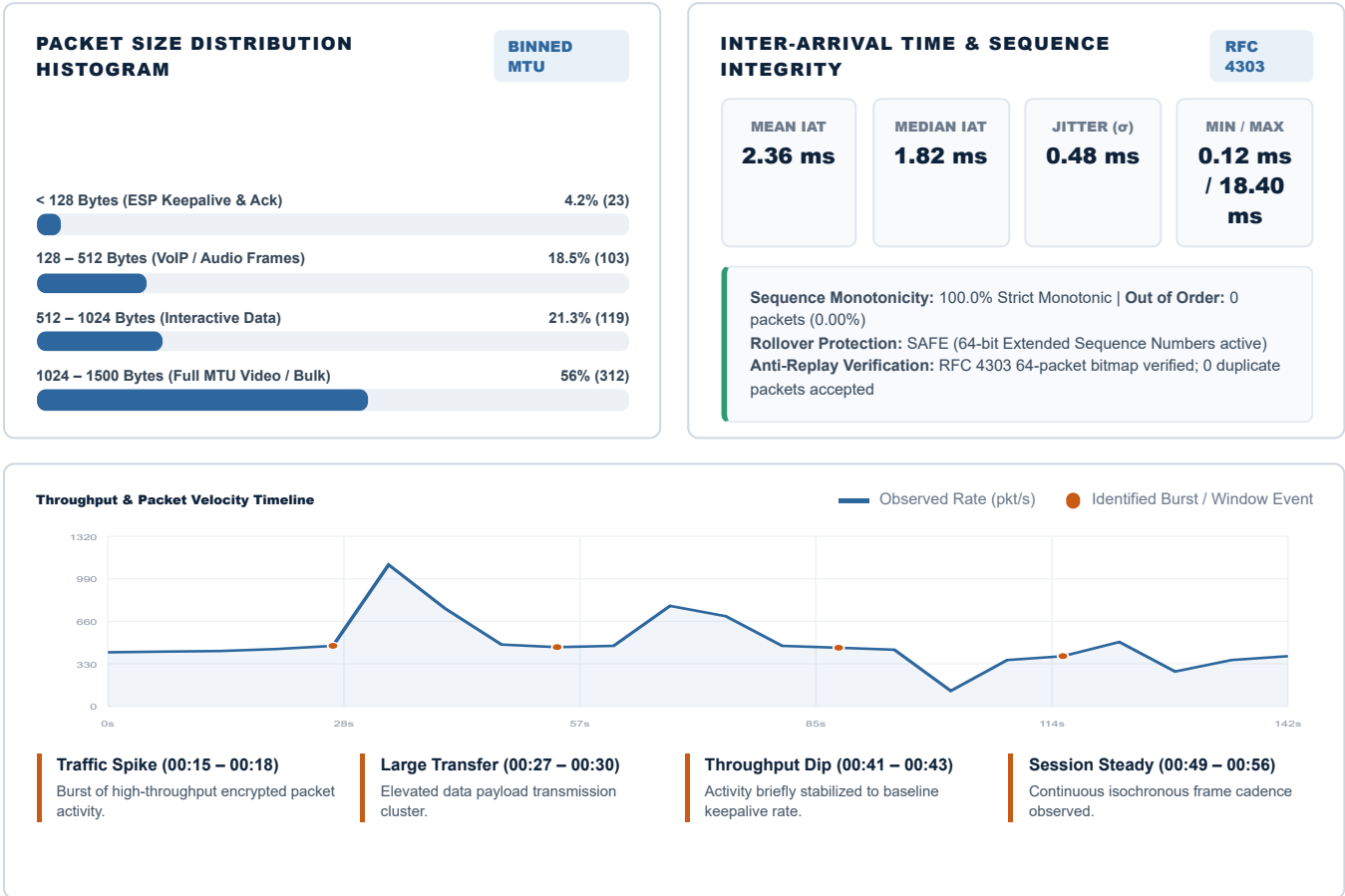
DATA-PLANE TELEMETRY & AI/ML PAYLOAD CLASSIFICATION

Data-Plane Stream Telemetry & Inference

Packet size distribution histograms, timing analysis, sequence progression, and inner encrypted traffic classification

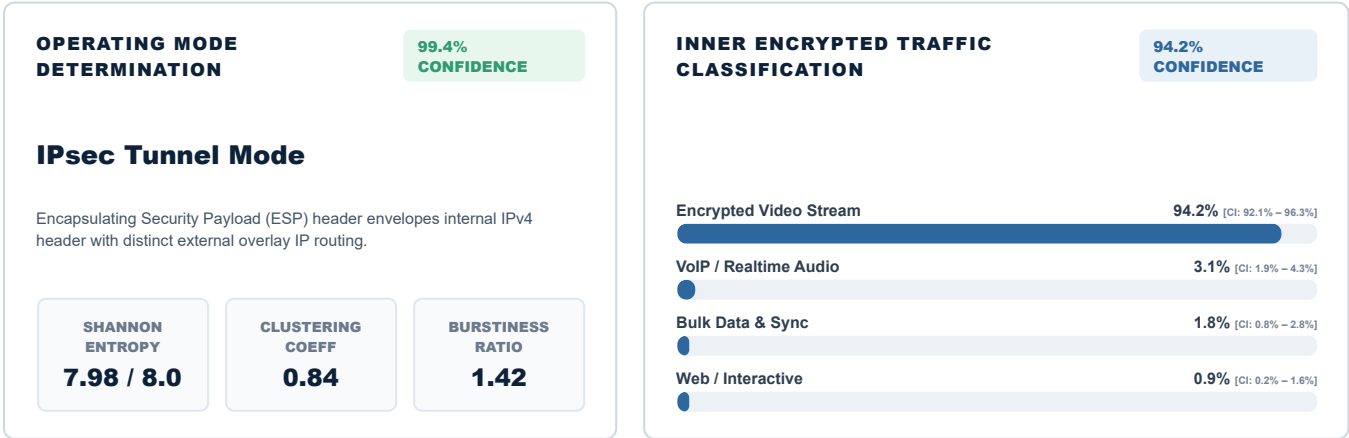
Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression



Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals





ENGINEERING REMEDIATION & RAW FORENSIC EVIDENCE

Implementation Patches & Trace Index

Actionable gateway configuration blocks ready for deployment, and raw packet trace appendix with flow JSON dumps

Section 5: Implementation Remediation Patches

Actionable configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

```
# 1. StrongSwan Gateway Patch (/etc/swanctl/conf.d/apex_hardened.conf)
# /etc/swanctl/conf.d/apex_hardened.conf
connections {
  apex-ipsec-hardened {
    version = 2
    local_addrs = 100.119.32.83
    remote_addrs = 100.127.207.119
    proposals = aes256gcm16-ecp256-sha256

    children {
      apex-child {
        # STRICT MANDATE: Enforce DH Group 19 (ECP-256) on Child SA for Perfect Forward Secrecy
        esp_proposals = aes256gcm16-ecp256!
        replay_window = 64
        life_time = 3600s
        life_bytes = 1000000000
        dpd_action = restart
      }
    }
  }
}
```

```
! 2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX_IKE_PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ipsec transform-set APEX_ESP_SET esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX_IPSEC_PROFILE
  set transform-set APEX_ESP_SET
  set pfs group19
  set security-association replay window-size 64
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS: `$ swanctl --load-all && swanctl --initiate --child apex-child` `$ swanctl --list-sas | grep -E 'AES_GCM|ECP_256'` `$ ip xfrm`

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#1	+0.000s	UDP 500	IKE_SA_INIT (Req)	Initiator=0x8b14e9f2...	384 B	DH Group 19 proposal exchange
#2	+0.014s	UDP 500	IKE_SA_INIT (Resp)	Responder=0x4a7c10b...	384 B	DH public key confirmation
#3	+0.028s	UDP 4500	IKE_AUTH (Req)	Initiator=0x8b14e9f2...	448 B	PSK mutual identity authentication
#4	+0.042s	UDP 4500	IKE_AUTH (Resp)	Responder=0x4a7c10b...	448 B	Child SA creation and Traffic Selectors

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SIEM COMPATIBLE

```
{
  "session_id": "AV-20260916-TECH-07BD",
  "protocol": "IPsec ESP / IKEv2",
  "endpoints": {
    "initiator": "100.119.32.83",
    "responder": "100.127.207.119"
  },
  "ike_transforms": {
    "encryption": "AES-256-GCM",
```

OVERALL TECHNICAL VERDICT

The analyzed IPsec session demonstrates hardened packet transport with AES-256-GCM encryption and strictly monotonic sequence progression. Implementing the provided swanctl / Cisco PFS remediation configuration will upgrade forward secrecy to meet NSA CNSA 2.0 and NIST SP 800-77 requirements.